

D651

Foundations of Design

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A design team at a grocery-delivery startup that just opened a new warehouse has interviewed customers and written a clear problem statement. The CTO loves whiteboards. What is the next stage of design thinking?

- A. Prototype
- B. Test
- C. Empathize again
- D. Ideate

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A team is building a paper mockup to show users. Which stage came just before this one?

- A. Define
- B. Ideate
- C. Test
- D. Empathize

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A team designing a medication-reminder app for older adults, at a company with a Seattle office, asks users, “What features do you want?” Most say “bigger buttons.” What is the best empathize practice?

- A. Watch users manage their medications at home to find unstated needs and workarounds.
- B. Build bigger buttons right away.
- C. Survey the company’s own staff instead.
- D. Users always know exactly what they need.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

After research for a city bike-share program that recently repainted its bikes, which is the strongest point-of-view statement?

- A. Riders like the new paint colors.
- B. We need an app with a live dock map.
- C. Make bike share better.
- D. One-way commuters need a reliable way to find open docks near work because arriving to full docks makes them late.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

In an ideation session at a furniture retailer with 18 stores, a senior manager critiques each idea as soon as it is offered. The session starts at 9 a.m. What is the main problem?

- A. The session is too early in the day.
- B. Senior managers must never attend ideation.
- C. Judging too early; ideas should be generated freely before any are narrowed.
- D. Too many ideas are being produced.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

After generating 80 ideas, a team clusters them and votes to pick three to prototype. What mode of thinking is this?

- A. Empathize
- B. Prototype
- C. Convergent
- D. Divergent

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A team at a bank wants to test a new account-opening flow and has one week. The bank's lobby was renovated last year. What is the best approach?

- A. Fully code the flow first, then test.
- B. Test a paper or simple clickable prototype with about five users, then revise.
- C. Skip testing and launch.
- D. Test the flow in the renovated lobby only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A hospital that recently won a design award wants to redesign the whole patient journey — check-in staff roles, the waiting process, billing, and follow-up calls — across people and systems. Which design field fits?

- A. Service design
- B. Product design
- C. UI design
- D. Graphic design, since the hospital won a design award

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A designer’s work is choosing button styles, colors, and fonts for a mobile app’s screens. Which field is this?

- A. UI (visual interface) design
- B. Industrial product design
- C. Service design
- D. UX research

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A transit agency that recently updated its logo is designing a ticket kiosk. The team has tested it only with average weekday commuters. What would make the design more equitable?

- A. Test with more average commuters.
- B. Make the logo on the kiosk larger.
- C. Design for the average user; edge cases can never be served.
- D. Include wheelchair users, low-vision riders, and non-English speakers in research and testing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A team at a sporting-goods chain that sponsors a city marathon designs a smart running shoe that testers loved. Each pair costs \$900 to make and would sell for \$150. Which lens fails?

- A. Marathon fit
- B. Desirability
- C. Viability
- D. Feasibility

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Users want a product and it would be profitable, but the battery technology it needs doesn't exist yet. Which lens fails?

- A. None; if users want it, build it.
- B. Feasibility
- C. Viability
- D. Desirability

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A design team at a grocery-delivery startup ~~that just opened a new warehouse~~ has interviewed customers and written a clear problem statement. ~~The CTO loves whiteboards.~~ What is the next stage of design thinking?

Target word: next stage **Decides:** Empathize and Define are done.

A	Dropped element	Skips generating ideas to choose from.
B	Confusion set	Nothing exists to test yet.
C	Inverted operation	Goes backward without a reason.
D	Correct	Generating ideas follows the defined problem.

2. Answer B Concept 1 • transfer

A team is building a paper mockup to show users. Which stage came just before this one?

Target word: prior stage **Decides:** Building a mockup is Prototype.

A	Dropped element	Define comes two stages earlier.
B	Correct	Ideate comes right before Prototype.
C	Inverted operation	Test comes after Prototype.
D	Confusion set	That is the first stage.

3. Answer A Concept 2

A team designing a medication-reminder app for older adults, ~~at a company with a Seattle office,~~ asks users, “What features do you want?” Most say “bigger buttons.” What is the best empathize practice?

Target word: better empathy method **Decides:** Stated wants miss unstated needs.

A	Correct	Observation in context reveals what users don’t say.
B	Dropped element	Skips deeper research.
C	Adjacent category	Wrong users.
D	Absolute language	Users often can’t state underlying needs.

4. Answer D Concept 3

After research for a city bike-share program ~~that recently repainted its bikes,~~ which is the strongest point-of-view statement?

Target word: best POV **Decides:** User + need + insight, no solution.

A	Camouflage	Not a need from the research.
B	Adjacent category	A solution, not a POV.
C	Dropped element	No user, need, or insight.
D	Correct	Specific user, need, and insight.

5. Answer C Concept 4

In an ideation session at a furniture retailer ~~with 18 stores~~, a senior manager critiques each idea as soon as it is offered. ~~The session starts at 9 a.m.~~ What is the main problem?

Target word: ideation flaw **Decides:** Judging during divergent thinking.

A	Camouflage	Timing isn't the problem.
B	Absolute language	They can attend if they defer judgment.
C	Correct	Early criticism shuts down idea flow.
D	Inverted operation	Quantity is the goal at this point.

6. Answer C Concept 4 · transfer

After generating 80 ideas, a team clusters them and votes to pick three to prototype. What mode of thinking is this?

Target word: name the mode **Decides:** Narrowing.

A	Confusion set	That is user research.
B	Adjacent category	Prototyping comes after choosing.
C	Correct	Narrowing many options to a few.
D	Inverted operation	Divergent generates; this narrows.

7. Answer B Concept 5

A team at a bank wants to test a new account-opening flow and has one week. ~~The bank's lobby was renovated last year.~~ What is the best approach?

Target word: prototype approach **Decides:** Little time, untested idea.

A	Inverted operation	Too slow and costly for early learning.
B	Correct	Fast, cheap learning before building.
C	Dropped element	No learning at all.
D	Camouflage	The lobby doesn't decide the method.

8. Answer A Concept 6

A hospital ~~that recently won a design award~~ wants to redesign the whole patient journey — check-in staff roles, the waiting process, billing, and follow-up calls — across people and systems. Which design field fits?

Target word: design field **Decides:** Journey across people, process, and touchpoints.

A	Correct	Covers end-to-end journeys across people and systems.
B	Confusion set	Focuses on a single product, not the whole service.
C	Adjacent category	UI covers screens, not staff roles or processes.
D	Camouflage	The award doesn't define the scope.

9. Answer A Concept 6 • transfer

A designer's work is choosing button styles, colors, and fonts for a mobile app's screens. Which field is this?

Target word: name the field **Decides:** Screen-level look and controls.

A	Correct	Screen appearance and controls.
B	Confusion set	Physical objects.
C	Inverted operation	Far broader than screens.
D	Adjacent category	Research studies users; it doesn't style screens.

10. Answer D Concept 7

A transit agency ~~that recently updated its logo~~ is designing a ticket kiosk. The team has tested it only with average weekday commuters. What would make the design more equitable?

Target word: equitable step **Decides:** Only average users were involved.

A	Dropped element	Still excludes those most likely to struggle.
B	Camouflage	Branding isn't equity.
C	Absolute language	Inclusive design often helps everyone.
D	Correct	Designing with people at the margins widens access.

11. Answer C Concept 8

A team at a sporting-goods chain ~~that sponsors a city marathon~~ designs a smart running shoe that testers loved. Each pair costs \$900 to make and would sell for \$150. Which lens fails?

Target word: failed lens **Decides:** Cost far above price.

A	Camouflage	Not a design lens.
B	Inverted operation	Testers loved it.
C	Correct	It can't make money: \$900 cost vs. \$150 price.
D	Confusion set	It can be built; it just costs too much.

12. Answer B Concept 8 • transfer

Users want a product and it would be profitable, but the battery technology it needs doesn't exist yet. Which lens fails?

Target word: failed lens **Decides:** Can't be built.

A	Absolute language	All three lenses must hold.
B	Correct	The technology isn't available.
C	Confusion set	It would be profitable.
D	Inverted operation	Users want it.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Design thinking stages	Empathize → Define → Ideate → Prototype → Test, looping back as needed.	Skipping from a problem statement straight to building.	“What comes next?”
2	Empathize through observation	Observing users in context uncovers unstated needs and workarounds that direct questions miss.	Asking users what features they want and building that.	A team relying on what users say they want.
3	Point-of-view statement	A POV names a specific user, a need (verb), and an insight (why), with no solution inside.	Writing a solution as the problem.	Choosing the best problem statement.
4	Divergent then convergent	Ideation first generates many ideas without judgment (divergent), then narrows them (convergent).	Critiquing ideas as they are offered.	Judging during brainstorming, or voting to narrow.
5	Low-fidelity prototyping	Cheap, fast prototypes tested with a few users early let teams learn before investing.	Building the full product before testing.	Limited time and an untested idea.
6	Design fields	Service design = whole journeys across people, process, and touchpoints; UX = a user’s experience with a product; UI/visual = screen look and controls; product design = the product itself.	Calling journey-wide redesign UI or graphic work.	The scope of what is being designed.
7	Equitable design	Involve people at the margins — disabilities, languages, low access — in research and testing; designing for them often helps everyone.	Designing for the “average” user.	Who the team tested with.
8	Desirability, feasibility, viability	Desirable = people want it; feasible = it can be built; viable = it can sustain a business.	Mixing up feasibility and viability.	Cost, technology, or user demand problems.

Course scope (public description): Foundations of Design introduces students to foundational theories and concepts of design. Students will develop a comprehensive knowledge of design thinking and its implications as a problem-solving approach to identifying user needs and solving business problems. Various types of design fields are introduced including product design, experience design, service design, visual and graphic design, and user experience design. Equitable design considerations are explored as students gain knowledge of the design process.